

Generalized path-connectedness in proximity spaces

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L1 - TRACTABLE

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Topology

ORIGINAL PROBLEM STATEMENT

Let δ be a proximity.

A set A is connected regarding δ iff $\forall X, Y \in \mathcal{P}A \setminus \{\emptyset\} : (X \cup Y = A \Rightarrow X \delta Y)$.

Conjecture The following statements are equivalent for every endofunctor μ and a set U :

- U is connected regarding μ .
- For every $a, b \in U$ there exists a totally ordered set $P \subseteq U$ such that $\min P = a$, $\max P = b$, and for every partition $\{X, Y\}$ of P into two sets X, Y such that $\forall x \in X, y \in Y : x < y$, we have $X [\mu]^* Y$.

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>